

Structured Query Language (SQL)

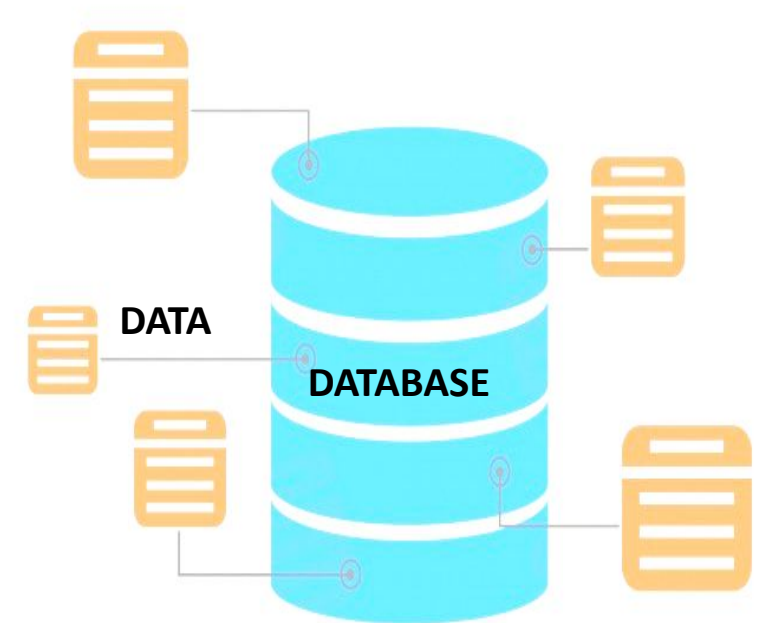
Introduction to SQL

DATABASE

A database is an organized collection of data.

For Example:

A university database organizes the data about students, faculty, admin staff, etc.



DBMS(Data Base Management System)

The software which is used to manage databases is called Database Management System. DBMS allows users to create ,read , update and delete data in database.

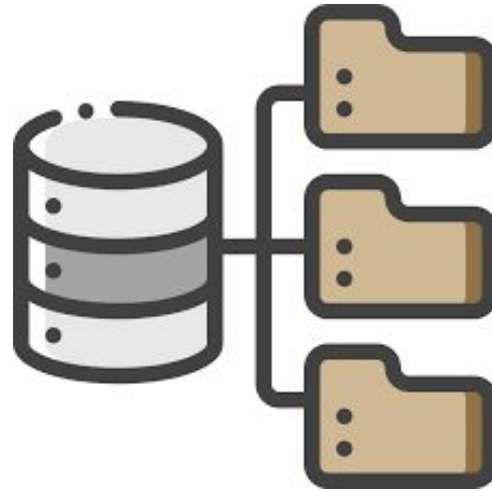
For example:

MySQL, Oracle, Microsoft SQL server, PostgreSQL are the popular commercial DBMS used in different applications.



WHY DBMS?

- Data Storage
- Data abstraction
- Controls data redundancy
- Multi - user access and views
- Security



DATA MODELS

- Data models define how the logical structure of a database is modelled.
- Data models define how data is connected to each other ,how they are processed and stored inside the system.

Hierarchical Model

Network Model

Entity Relationship Model

Relational Model

No SQL

RELATIONAL MODEL

In this model , the data is organized in the form of two dimensional table. That means data is stored in the form of rows and columns.

For example:

The table here shows “STUDENT DETAILS” with attributes such as Stu_Id, Name and Branch which has 5 records.

Stu_Id	Name	Branch
201	John	CSE
202	Mary	ECE
203	Britto	IS
204	Harry	IT
205	Steve	ME

RDBMS

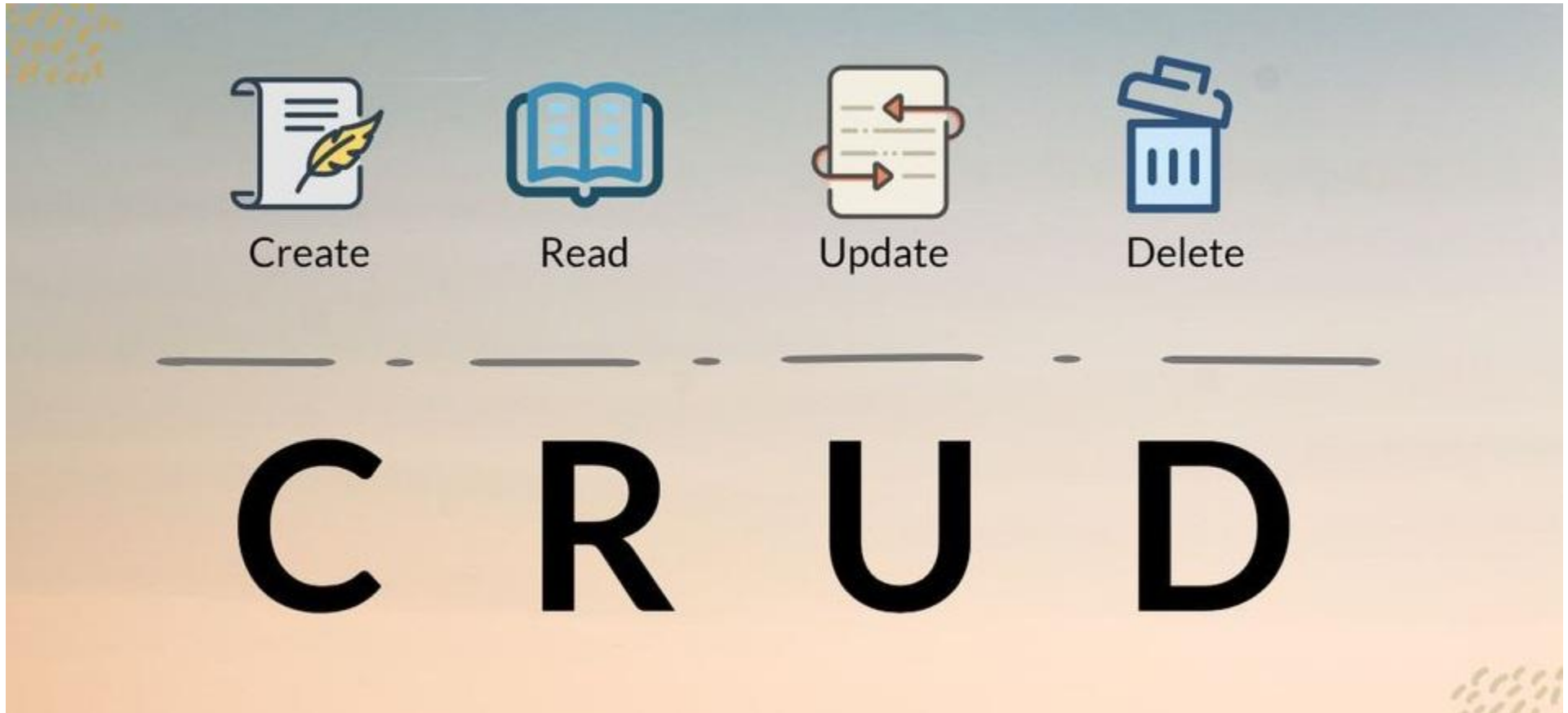
RDBMS is a program which is used to manage Relational databases in which data is stored in the form of rows and columns which can be easily retrieved ,managed and updated.



SQL

- SQL stands for Structured Query Language.
- SQL is a database programming language designed for the retrieval and management of data in a relational database.
- All relational database management systems like MySQL ,Oracle , MS Access and SQL server uses SQL as their standard database language.

CRUD OPERATIONS



End of Introduction to SQL